Dosimetry, half-life period, χ^2-Test

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Caesium-137 was used as the source of γ-rays in this experiment. The equivalent dose rate for a distance of 5 cm between the source and the counter is estimated to be $\dot{H}_{5\text{cm}} = (9.88 \pm 0.29)\ \frac{\mu\text{Sv}}{\text{h}}$ while it decreases to $\dot{H}_{1\text{m}} = (0.152 \pm 0.027)\ \frac{\mu\text{Sv}}{\text{h}}$ for a distance of a meter. The average equivalent dose for the experiment is $H_{\text{avg}} = (0.312 \pm 0.013)\ \mu\text{Sv}$. The activity of the Caesium-137 source at the time of the experiment is $A = 1392$ kBq. χ^2-tests showed that the radioactive decays measured with a gate-time of 1s and 10s were Poisson distributed. The measurements with the Geiger-Müller counter were executed at an operating voltage of $U = (560 \pm 4)$ V since it is part of the plateau voltage. The measurement of the zero-effect yields $Z = (28.2 \pm 2.4)$. The decay curve of Ba-137m was recorded and its half-life time is calculated to be $T_{\frac{1}{2}} = (153.6911 \pm 0.0098)$ s. The responsiveness of the counter is measured to be $(0.627 \pm 0.016)\%$ at a distance of 20 cm and $(0.702 \pm 0.023)\%$ at a distance of 40 cm from the source.

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1 Introduction

Radioactivity is described as the spontaneous decay of atom cores in which ionising particles or highly energetic rays are emitted. Basically there are three types of ionising radiation which are called α-, β- and γ-radiation. α-particles consist of two protons and two neutrons that escape the atom core by tunneling through its coulomb wall while β-particles are fast electrons or positrons that are emitted if β-decay occurs. γ-radiation, however, is not developed by a decay but rather is an accompanying effect of a decay wherein an excited atom core relaxes to its ground state, thus releasing a photon of high energy in MeV range. Such photons are called γ-rays. Since all these radiations are dangerous to all living beings, it is necessary to understand their properties as well as devise ways to detect them reliably. In this work, we describe ways to systematically measure the quantifiers of a radioactive decay. We also investigate the distribution trends for the number of recorded decay events, to see if the theoretical assumptions behind the "law of radioactivity" are being abided to.

2 Dosimetry and radiation exposure

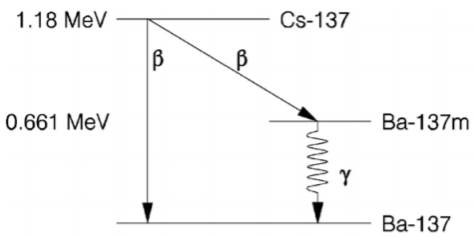

Figure 1: Decay scheme of Cs-137[2]. It can be seen that Cs-137 decays either directly into Ba-137 or first into Ba-137m and then into its ground state. In the second process the gamma ray is produced.

In this part, the γ-radiation of Cs-137, which emits photons with an energy of 661 keV[2], is observed. Cs-atoms decay with a half-life time of 30.2 years into Ba-137. Sometimes they decay into the meta-stable Ba-137m. This is an excited state of Barium that emits the said photon when relaxing into the ground state (figure 1).

To measure this radiation the NEVA-dosimeter was

used. It was set to short-term measurement. This dosimeter showed the equivalent dose per hour. The γ-ray source was inside of a lead cube. It had a hole which was closed with a removable plug. By removing the plug a collimated γ-ray beam was being emitted that could be measured. At first the dosimeter was held right above the hole (roughly 5cm away from the source). The same procedure was applied in 1m distance. Furthermore another dosimeter was used in 1m distance for the whole duration of the experiment to track the average exposure to radioactivity.

At the distance of 5 cm an equivalent dose per hour of

$$\dot{H}_{5\text{cm}} = (9.88 \pm 0.29)\,\frac{\mu\text{Sv}}{\text{h}}$$

was recorded. Multiplied by 4h hours yields the dose that students would be exposed on that day if the source was 5 cm away the whole time:

$$H_{5\text{cm}} = (39.5 \pm 1.2)\,\mu\text{Sv}$$

This value seems to be very large considering that it is not recommended to exceed the exposure of 5 mSv per year. Extrapolated to one year one would be subjected to $H_{5\text{cm}} = (86.5 \pm 2.5)\text{mSv}$. Of course this would be only the case if the whole work was done 5 cm away from the source.

A much smaller radiation was registered in 1 m range:

$$\dot{H}_{1\text{m}} = (0.152 \pm 0.027)\frac{\mu\text{Sv}}{\text{h}}$$

Again extrapolating this result for 4 hours gives:

$$H_{1\text{m}} = (0.61 \pm 0.11)\,\mu\text{Sv}$$

Comparing it to the recommended 5 mSv per year it is only 0.012%. Due to the measurement being disturbed by the background radiation we have to subtract the latter from the measurement in 1 m distance. For this correction a measurement was done every 15 minutes in 1 m distance and the average value was taken. The result for this is $\dot{H}_{\text{avg}} = (0.1083 \pm 0.0046)\frac{\mu\text{Sv}}{\text{h}}$. It can be seen that the radiation exposure is much smaller. Furthermore someone needs to work for 32968 hours (around 45 months) in 1 m distance to the source to exceed the recommended value. Evidently, it is impossible to be exposed to a radiation of 5 mSv per year in a working environment like this.

3 Activity

We would require the information about the present activity of Cs-137 for the upcoming experiments. In order for that, it is necessary to describe the radioactive decay at first.

If N_0 is the number of particles of the material before the decay, the amount of not decayed particles after the time t is given by

$$N(t) = N_0 \cdot \exp(-\lambda t) \tag{1}$$

The equation $N(t) = \frac{1}{2} \cdot N_0$ delivers the half-life time $T_{\frac{1}{2}}$ at which point the original particle number has been reduced to half:

$$T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}. \tag{2}$$

The activity A is defined as the number of decays in a time frame dt, which equals the product of the decay constant λ with the particle number $N(t)$ from the law of radioactivity (equation 1).

$$A = \lambda \cdot N(t) = -\frac{\mathrm{d}N(t)}{\mathrm{d}t} \tag{3}$$

λ can also be expressed as the probability of a decay in the time frame. Putting all these three expressions together gives the equation that was used to get the activity during the procedure

$$A(t) = A_0 \cdot \exp\left(-\ln(2) \cdot \frac{t}{T_{\frac{1}{2}}}\right). \tag{4}$$

When the object was delivered to the laboratory on the 21st August 1978 it had an activity of $A_0 = 3.7\,\text{MBq}$. The half-life time is 30.2 years. t here is the difference between the delivery date and the date of the experiment (42.6 years). Alltogher this yields:

$$A(40.62\text{ a}) = 1392\text{ kBq}$$

4 Geiger counter characteristics

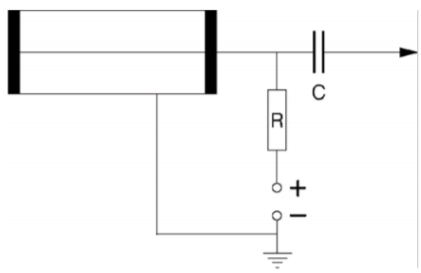

Figure 2: Schematic setup of a Geiger counter[2]. High energy particles can be registered with its aid.

For this part a Geiger-Müller (GM) counter was used with its schematic setup illustrated in figure 2. It consists of a tube that is negatively charged and a positively charged, thin metal, wire which runs through the center of the tube making it similar to a cylinder capacitor. The tube is filled with a mix

of alcohol vapor and argon. The wire is connected to a high resistor. If a γ-ray enters the tube it ionises the particles of the gas. The negative particles travel much faster than the positive ions. Since the negative particles are attracted by the positive wire a voltage pulse can be measured at the resistor. This can only happen if a high enough operating voltage is reached. If it is not high enough, the ionised particles would be too slow to reach the wire and would rather recombine with the ions on their way. This way no voltage would be registered. However if the voltage is high enough the electrons would have an energy large enough to produce secondary ionisations, thus increasing the voltage pulse significantly.

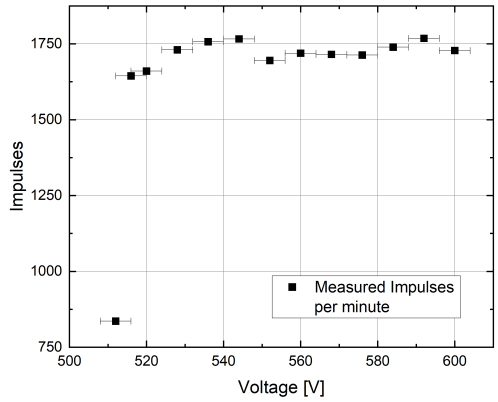

Figure 3: Impulses measured with the Geiger counter in dependance of the voltage to find its operating threshold.

The voltage, at which the counter starts measuring impulses, is $U = (512 \pm 4)$ V. The error comes from the fluctuations of the voltage meter. It is noticeable that in the first few measurements an increase is to be seen. This is explained by the decreasing recombinations in the gas mix. The voltage was continuously increased to a point that delivered nearly the same amount of measured impulses, the so-called plateau voltage. We decided to use $U = (560 \pm 4)$, which belongs to the plateau region, as our operating voltage. With that voltage all following measurements were made.

5 Zero-effect

The zero-effect describes the radiation that occurs if there is no radiating material around. Background radiation is responsible for this. For later experiments it is wise to clear the the background from the data. It is therefore necessary to determine it first. The Geiger counter was placed at a location in the laboratory that was far enough from radioactive materials. Later on for comparison the counter was placed inside of a lead shielding. In both cases a series of measurements was taken with a gate time of 1 minute. Outside of the shield an average value of Impulses per minute of

$$N_{\text{out}} = 28.2 \pm 2.4$$

was measured while inside of it the value was

$$N_{\text{in}} = 15.8 \pm 1.3.$$

The errors are both standard errors. With a γ-ray reduction of 44,0% it is easy to see that the lead does indeed very well detain a big portion of the radiation.

6 Counting statistics

In this section we examine the decay rate for a tile coated with Uranium oxide. Time frames shorter than the half-life of the source would deliver a constant decay rate. Besides, the measurement should tend towards a Gaussian distribution as the number of observations rises. The goal is to see if the decay rate follows a Poisson distribution and if it tends towards Gaussian distribution for longer gate times. For this, the tile, alongside with a GM counter, was enclosed in a lead shielding and the number of decay events with a constant gate time was recorded. Since the lead shielding for this part was of a better quality than the one used in section 5, it is not necessary to clear the background radiation from the data.

6.1 Measurement series with 1s gate time

At first, the gate time was set to 1s and 1000 measurements were taken which can be seen in figure 4 or table 1, where the last two classes were combined. To figure out if these numbers follow a Poisson distribution, a χ^2-test is done. With an error of 5% a threshold of $\chi_0^2 = 7.815$[3] is set for the null hypothesis that the data follows such a distribution.

Number of decays	Frequency
0	462
1	372
2	135
3	26
4-∞	5

Table 1: Group classification of the decays in the 1s gate time. For this the classes 4 and 5 were combined in the class 4-∞ since the decays of both classes are < 5.

Since the number of the last two classes of 4 and 5 decays per second is less than 5 the classes had to be put together into the class $4-\infty$. The given data has $\chi^2 = 3,396$ with an average value of $\mu = (0.740 \pm 0.027)$ decays. The error is calculated through $\sigma_\mu = \frac{\sqrt{\mu}}{\sqrt{n}}$. As this χ^2 is much smaller than the threshold the null hypothesis is therefore right. The data and the decay rate does follow a Poisson distribution. If one wanted to say that the data is not subjected to a Poisson distribution an error of more than 30% would be necessary.

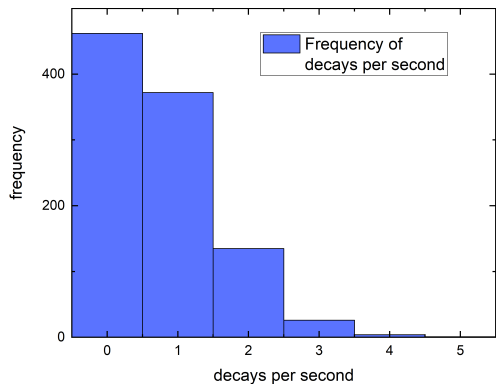

Figure 4: Histogram of the observed decays with a gate time of 1s.

6.2 Measurement series with 10s gate time

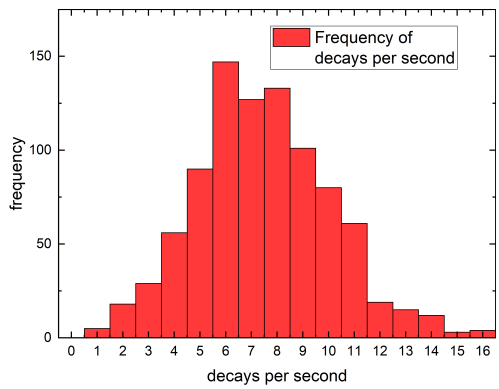

Figure 5: Histogram of the observed decays with a gate time of 10s.

Now the gate time was changed to 10 seconds and all in all 900 measurements were taken here. This way the average decay value is much higher (around ten times) than for 1s. The distribution should now get much closer to a Gaussian distribution. The data can be taken from table 2. Additionally the data in table 2 has been plotted and can be seen in the histogram (figure 5). As with the gate time of 1s, we now again formulate our null hypothesis to be that the data follows a Poisson distribution with an error of 5%. Looking up the threshold for this case delivers $\chi_0^2 = 22.362$ [3].

With an average value of $\mu = (7,469 \pm 0,091)$ decays a $\chi^2 = 19,774$ is calculated. Again the calculated χ^2 is smaller than the threshold. Therefore the null hypothesis must be accepted, meaning the data for this part of experiment is also Poisson distributed. If the data should still not follow a Poisson distribution one would have to accept an error of around 15%.

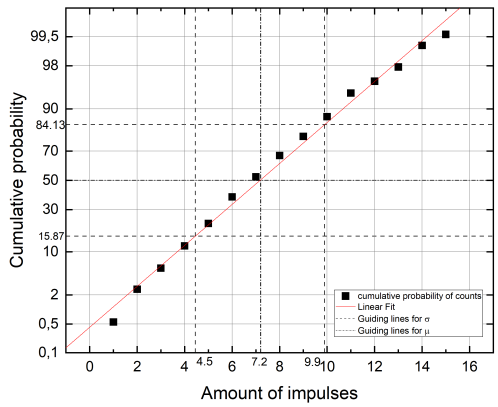

Figure 6: Cumulative probabilities in dependance of the amount of impulses for the measurement with 10s gate time. The data was fitted with a linear fit and guiding lines. The lines were placed at 84.13% and 15.87% and the associated value on the x-axis was read off. Moreover a guideline for the average value was drawn. With this the standard deviation was calculated.

To find out whether the data can be described with a normal distribution the cumulative probability is plotted and fitted linearly. This can be seen in figure 6. The average value is read off at 50%. Furthermore the values at 15.87% and 84.13% are taken from the graph to calculate the standard deviation. The results are:

$$\mu_{\text{Graph}} = (7.20 \pm 0.10)$$
$$\sigma_{\text{Graph}} = (2.70 \pm 0.14)$$

If we compare these results with the calculated ones $\mu = 7.469 \pm 0.091$ and $\sigma = \sqrt{\mu} = 2.73 \pm 0.30$ one can assume that the data is normally distributed. The χ^2-test, however, shows that they are Poisson distributed. In order to check the distribution definitively, another χ^2-test for a normal distribution is executed. The sample mean and the standard deviation of the data are used for this test. The null hypothesis is that the data follows the Gaussian distribution and the error is set again to 5% with 12 degrees of freedom resulting in a $\chi_0^2 = 21.026$[3]. Computing the χ^2 with the measured data gives $\chi^2 = 21.806$. Since the

Number of decays	Frequency
0-1	5
2	18
3	29
4	56
5	90
6	147
7	127
8	133
9	101
10	80
11	61
12	19
13	15
14	12
15-∞	7

Table 2: Group classification of the decays in the 10s gate time. For this the classes 15-16 and 0-1 were combined in the class 15-∞ since the decays of those classes are < 5.

calculated χ^2 is higher than the threshold the null hypothesis must be rejected meaning that the data does not follow the Gaussian distribution.

Both the χ^2-tests show that the data can not be described by the Gaussian distribution but rather by the Poisson distribution. These tests verify the Poisson distribution of the data definitively since the graphical method is only helpful for finding the qualitative answer for the distribution. The graphical method should only be used to approximate the mean value and standard deviation of the data.

7 Half-life-time of Ba-137m

The radioactive Cs-137 is actually a β-emitter. However, it can decay into the excited state of Ba-137 that on the other hand decays under emission of a γ-ray into its ground state. The preparation was chemically washed out of Cs-137. By doing so only Ba-137 alongside its excited state were seperated out. In general excited states are short-lived. The the gate-time was set to 10s.

The decay is described by an exponential function as seen in equation (1). This can also be seen in figure 7. The data is fitted with an exponential function with an additive constant for the zero-effect. The decay constant is derived from the parameters of the fit:

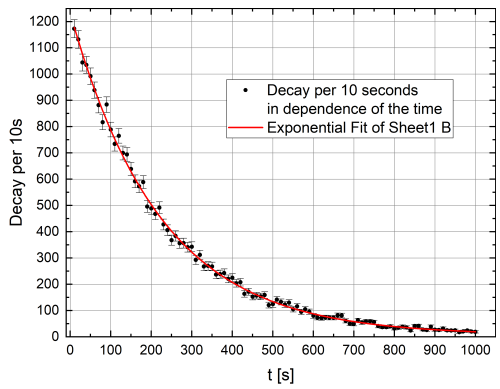

Figure 7: The recorded decays of Ba-137m are seen in this figure. The error bars are Poisson errors. The data was fit with an exponential function and with this the half-life time as well as the zero-effect was determined.

$$\lambda = (4.514 \pm 0.053) \cdot 10^{-3} \frac{1}{s}.$$

The zero-effect is determined the same way:

$$Z_{10s} = (6.1 \pm 1.6)$$

Projecting this value to one minute it seems to be larger than the one calculated in 5. An explanation is that the measurement was stopped slightly too early and therefore the remnant radiation from Ba-137m distorted the result.

Taking the result of λ and plugging it into equation 2 gives us the half-life time of Ba-137m.

$$T_{\frac{1}{2}} = (153.6911 \pm 0.0098)\ s$$

Comparing it to the literature value of 153.12 s[4] it can be seen that it does not quite match. However the values differ only by 0.37% so the result is absolutely plausible.

8 Responsiveness of the Geiger counter

The responsiveness Z gives information about the quality of the Geiger counter and is defined as the quotient between the real number of rays hitting the counter N_{re} and the number of measured rays N_{me}:

$$Z = \frac{N_{\text{me}}}{N_{\text{re}}} \tag{5}$$

In this case the Cs-137 source was taken out of its lead shielding and placed on a mount to ensure that the whole area of the counter is hit by the radiation being emitted. Both were placed on the same height. To get a better conclusion the measurement was made for two different distances, $d_1 = 20$ cm and $d_2 = 40$ cm, between the counter and the source. In both the cases a series of measurements was done for the number of measured impulses per minute and the average value was calculated. For the real number of rays hitting the sensory area F of the counter

$$N_{\mathrm{re}} = A \cdot 60\mathrm{s} \cdot \frac{F}{4\pi d^2} \tag{6}$$

is used. Here the determined activity A from 3 is plugged in. The results for both the distances are shown in table 3.

distance /m	impulses per minute	responsiveness
0.20 ± 0.01	1145 ± 24	$(0.627 \pm 0.016)\%$
0.40 ± 0.01	320.6 ± 7.4	$(0.702 \pm 0.023)\%$

Table 3: The table shows the calculated responsiveness. The result is corrected by the zero-effect.

Although the resulting Zs aren't too far from each other the gap is still noticeable even with their errors. One reason might be that the radiation distribution throughout the surrounding area is assumed to be equally distributed in the formula. But in the experiment the material was placed on a mount which does shield a part of the radiation. What also comes into play is the null effect that gets more significant the further away the counter is from the radiation source as it gets more difficult to distinguish between null radiation and radiation from the object. This gets even worse if the fact that not all the radiation hitting the sensor (due to the "dead-time" of the counter) is being measured for shorter distances, is also taken into account. But more important is the fact that in both the cases Z is very small (smaller than one percent). This means that only a fraction of the γ-rays create ion-pairs and interact with the gas.

9 Conclusion

To begin with, the radiation exposure was measured to be $H_{1\mathrm{m}} = (0.61 \pm 0.11)\ \mu\mathrm{Sv}$ from which it could be concluded that during the 4 hours of working-duration in the laboratory, only a harmless dose of radiation was taken in. This means a percentage exposure of 0.012% of the recommended maximum exposure per year. Following this, the activity of of the Cs-137 compound used in this experiment was calculated to be $A(40.62\,\mathrm{a}) = 1392\,\mathrm{kBq}$. Next, a tile painted on one of its faces with Uranium oxide was studied to see if its decay follows a Poisson distribution. In order to do so, a GM counter, with an operating voltage of $U = (560 \pm 4)$ V (which belongs to the characterisation plateau), was used. The GM counter alongside with the tile was enclosed in a lead shielding. The recording in this setup showed that the decay is indeed Poisson distributed. Besides, the half-life time of Ba-137m was estimated to be $T_{\frac{1}{2}} = (153.6911 \pm 0.0098)\,s$, which matches well with the given value of 153.12. The responsiveness of the GM tube was also studied to be $(0.627 \pm 0.016)\%$ at a distance of 0.20 m, and $0.702 \pm 0.023)\%$ at a distance of 0.40 m from the Caesium-137 source. Besides, null radiation detected by this GM counter was determined to be $N_{\mathrm{out}} = 28.2 \pm 2.4\ \frac{1}{\mathrm{min}}$ and $N_{\mathrm{in}} = 15.8 \pm 1.3\ \frac{1}{\mathrm{min}}$ with and without a lead shielding around the counter, respectively. These null values were used to correct the data collected by this GM counter, wherever applicable.

References

[1] Dieter Meschede, Gerthsen Physik, 25th edition, Springer-Verlag, 2015

[2] Instruction manual for the expermient, `https://www.physik.uni-wuerzburg.de/fileadmin/physik-grundpraktikum/_imported/fileadmin/11016800/Modulbeschreibungen/C-Modul/Versuch_46_ueberarbeitet.pdf` , last accessed on 19.03.2021

[3] `https://people.richland.edu/james/lecture/m170/tbl-chi.html`, last accessed on 19.03.2021

[4] `https://www.nndc.bnl.gov/`, last accessed on 20.03.2021